

A Self-Adaptive Traffic Light Control System Based on Speed of Vehicles

Jinyang Li* Yuanrui Zhang* Yixiang Chen*[†]

*School of Computer Science and Software Engineering,
MOE Research Center for Software/Hardware Co-Design Engineering,
East China Normal University, Shanghai 200062

[†]Corresponding author. Email: yxchen@sei.ecnu.edu.cn

Abstract—Intelligent transportation is a typical case of cyber-physical system (CPS). Due to the rapid increasing of the number of vehicles in city, problems caused by vehicles, like congestion and environment pollution, are becoming more and more serious. Traffic light control system is often used to control the vehicles passing for a solution of the congestion in the city. Present control systems used are normally assigned as to be static, i.e., traffic light signal changes in a static way. The aim of this paper is to propose a dynamical traffic light control system, i.e., change the traffic light signals in real time following the speed of vehicles. This system is an instance of V2I (Vehicle to Infrastructure) communication model, realizing data transmission between vehicles and traffic lights. Vehicles send speed messages to the traffic light when passing an intersection, then the traffic light analyzes the information and adjusts the signal time in real time. Each traffic light in each direction has a control strategy of itself without the orthogonal requirement. Therefore, the traffic light is a kind of cyber-physical system. This traffic light control system can maximize the number of vehicles passing intersection, and as a result, minimize the congestion and pollution. A traffic light control algorithm based on speed of vehicles and its simulation are presented. The safety and liveness of this control system are discussed too.

Index Terms—Intelligent transportation; Traffic light control system; Algorithm and Simulation

I. INTRODUCTION

Transportation is closely related to people's daily life, and people are strongly dependent on transportation. Intelligent transportation system (ITS) [1], [2] has been developing, but in most countries and cities, the speed of traffic construction is far behind the speed of the increase of the number of motor vehicles. The imbalance between these two speeds leads to traffic congestion, traffic accidents, environmental pollution and other issues. One of the most serious problem is the traffic congestion. For a long time, many researchers have been committed to solve this problem. It is well known that if the intersection of roads are occupied by vehicles then the congestion occurs. In order to decrease the congestion, a great amount of traffic light control systems are used in the city, town and village to control the vehicles passing in order.

At present, most used traffic light control systems are static, i.e., traffic light signal changes in the designed time periods rather than following the real situation of vehicles passing intersections. The designed time periods are sometimes given as statistical results based on the the historical data. Indeed, one uses sampling method to survey the information of traffic

intersection, then uses the statistical theory to calculate the delay time of the traffic lights in two directions, and sets the time in advance [3]. But, in practice, the number and speed of vehicles in intersections are dynamic, there is a big difference between the traffic conditions in different time. So the traffic lights with fixed time can not match the need of actual intersection. Therefore, one has to find an intelligent traffic light control system, which can automatically adjust the time span of the traffic signal according to the number and speed of the cars passing the intersection in real time. This kind of traffic light control system can maximize the number of vehicles passing the intersection and minimize the number of stranded vehicles, as a result, can effectively alleviate traffic congestion. This kind of traffic light control system is called a self-adaptive traffic light control system (SATL control system, shortly).

SCOOT [4] in England and SCATS [5] in Australia are two famous and successful SATL control systems in the world. [6] and [7] develop a class of SATL control systems based on wireless sensor network. [8] proposes a SATL control system based on fuzzy control algorithm. These mentioned SATL systems collect the road information and then transfer to a center controller in the back end. This kind of handling pattern of messages may impact the real-time performance in some ways. [9] introduced a SATL control system for a single intersection. This system uses photoelectric switch to count the number of the cars and change the signal time according to the traffic volume. Most of the systems mentioned above adjust the signal time according to the number of the cars, and control the traffic light change based on the orthogonality of the traffic lights in two directions.

The aim of this paper is to propose a novel SATL control system. In which traffic light signal changes in real time following the speed of vehicles. It is an instance of V2I (Vehicle to Infrastructure) communication model realizing data transmission between vehicles and traffic lights. Vehicles send speed messages to the traffic lights when passing an intersection, then the traffic light analyzes the information and adjust the signal time in real time. This kind of SATL control system has another feature: non-orthogonality. That is, each traffic light in each direction changes the traffic light signals based on the speed of vehicles without the orthogonal requirement. So it is necessary to ensure the safety and liveness

of the system. This traffic light control system can maximize the number of vehicles passing intersection, and as a result, minimize the congestion and pollution. In this paper, a traffic light control algorithm based on speed of vehicles and its simulation verifying its correctness are presented and the safety and liveness of this control system are analysed too.

This paper is organized as follows: Section II is about the system design, which describes the scene the system applies to, the communication mode, and the system architecture. Section III gives a detailed description of how the system is implemented, focusing on the control algorithm and its simulation. Section IV describes the related work. Section V sums up the paper work, and gives the next work about this system.

II. SYSTEM DESIGN

A. Scene Description

This system is applied to a single intersection. To reduce system complexity and make it easy to implement, we will not consider the case of a vehicle turning left or right, that is, we assume that every car runs straightly.

This traffic light control system can receive two speeds of a vehicle in every direction: one is when the vehicle passing the intersection, the other is the speed of the vehicle on the next section after the vehicle passes the intersection. Thus the system can judge the traffic condition, and then gives instructions to the traffic light, like extending the length of the signal time, switching the traffic light, and so on. The role played by this system is equivalent to an experienced traffic police, who can direct traffic according to the current situation and ease the congestion when it occurs.

B. Communication Mode

Vehicle to Infrastructure (V2I) [10], [11] is a concept from Internet of Vehicles (IoV) [12]. V2I means the communication between vehicles and the road infrastructure. Road infrastructure includes road, street lights, traffic lights, electronic barricades, etc. The system we design is a typical system of V2I mode. We can see from Fig. 1 that the system uses the star network topology, i.e., every vehicle communicates with the traffic light in the intersection. There is no interaction between car and car. Cars just send messages to the traffic light, and the traffic receive all these messages.

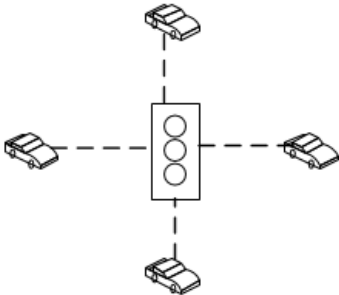


Fig. 1: System Network Topology

C. Modular Architecture

Two main parts of the system are the traffic light and the vehicles. Every part can be divided into different modules according to different functions. The modular architecture is shown as Fig. 2.

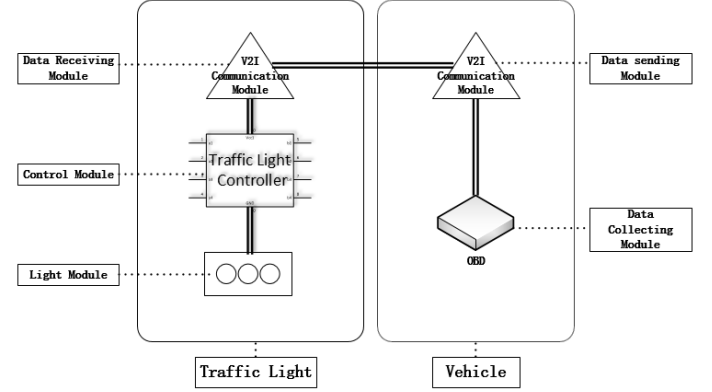


Fig. 2: Modular Architecture

The traffic light part includes data receiving module, control module and light module. Data receiving module is responsible for receiving data sent over by the vehicles. Control module, or shortly, the controller, is in charge of data processing and instructions sending. And the light module is for displaying. On the vehicles part, there are data collecting module and data sending module for collecting and sending speed, location, direction and other messages of the vehicles.

Under this architecture, the system activity can be described as follow: The data collecting module keeps collecting vehicle messages, including speed, location, direction, and so on. When the car runs into the system scope (the maximum data transmission distance supported by system), the data sending module sends these messages to the data receiving module on the traffic light. The traffic light transfers the received data to the controller. The controller judges the condition of the intersection and take actions to give proper instructions according to the algorithm. At last, the light module extends the light time or switches the light. This series of activities are iterative.

III. SYSTEM IMPLEMENTATION

A. Data Collecting Module

The data collecting and sending device in this system is an On-Board Diagnostic (OBD) device [13]. It is invented by National Trusted Embedded Software Engineering Technology Research Center. This device can automatically fit a variety of vehicle buses and finish data collection. We can directly connect it to the OBD interface of the car. And it can send the data away after some configurations.

B. V2I Communication Module

1) *Communication Device:* We use ZigBee [14] module as the V2I communication module in this system. ZigBee is

a short-range, low-power, low-cost wireless communication technology. ZigBee can spread 30-50 meters away in the room, and is able to reach about 100 meters outdoors. ZigBee has strong self-organizing, strong self-healing capabilities, big network capacity and good communication reliability, so it can meet the system requirements.

2) *Data Format*: The format of the data that travel from vehicles to the traffic light is defined as follow and is shown as Table I. A datum has 19 bytes, $Data[0]$ (the first byte) to $Data[18]$ (the last byte). $Data[0]$ uses a fixed value 0x55, signifying the beginning of the data. The next two bytes is the vehicle ID. Then is some information of the vehicle. 2 bytes for speed, 4 bytes for latitude, 4 bytes for longitude, and 4 bytes for driving direction. The speed collected by OBD is integer, so it can be directly converted to hexadecimal. Latitude and longitude are decimal with 5 decimal places, so we have to convert them to integer first and then convert them to hexadecimal. Driving direction is represented by azimuth angle, and we set due north to be 0 degree, due east to be 90 degrees, and so on. Also, we convert them to hexadecimal. We use $Data[17][18]$ as the end of a datum, using the fixed value 0x0D 0xAA. We use two bytes here rather than one, because all the data are random, 0x0D and 0xAA may occur and disturb judgment, two bytes can reduce the misjudgement rate.

Byte Number	Length(byte)	Content
0	1	Head of data
1	2	Vehicle ID
3	2	speed
5	4	Latitude
9	4	Longitude
13	4	Driving direction
17	2	End of data

TABLE I: Data format

The OBD sends data in the above format, and when the control module receives the data, it parses and decodes the data with the same logic and gets messages.

C. Control Module

1) *Controller*: Traffic light controller is the most important part of this system. The controller is responsible for executing the control algorithm and controlling the lights.

2) *Control Algorithm*: The main problem we need to solve is how to judge traffic condition according to speed information and to give correct instructions to lights.

In our previous work, a simple algorithm is proposed. We just consider the traffic light in the east-west (EW) direction, and the other is diametrically opposite to it. In other words, if the light in EW direction is green, then set the light in south-north (NS) direction to be red.

Now, we think about how to coordinate two traffic lights. The main of the control algorithm is: For both traffic lights in EW and NS directions, if no special circumstance occurs, base time for red and green lights is 60 seconds. We set two critical speeds V_{max} and V_{min} . Speed over V_{max} indicates that the road is clear. And the speed below V_{min} means

congestion. We control each traffic light according to speed and current light. If the current light is green, in two cases we need to extend the green time. One case is that this direction is clear, and the other direction is blocked. In this case, we extend the green time to avoid the vehicles in the blocked direction running into the intersection. Another case is that this direction does not have congestion and there are cars going to pass the intersection, while there is no car in the other direction. So we lengthen the green time to reduce waste of road resources. Once congestion occurs in this direction, whether in the intersection or the next section, turn the green light into red. If the current light is red, basic conditions are similar to conditions under green. Before taking actions to the lights, we need to check the remaining time and how long has the light been kept. If the remaining time is greater than 10 seconds, we don't extend the time. If the remaining time is less than 10 seconds, we don't shorten the time. Also we don't switch a light unless it has kept green or red for 30 seconds. This step is to avoid the waste of resources and lights switching too often. The specific algorithm is defined in Algorithm 1.

a) *Algorithm Analysis*: This algorithm applies to both EW direction and NS direction. In the algorithm, the subscript capital letter "T" means "This direction", while "O" means "Opposite direction". In EW direction, we consider about the cars coming from west to east, and in NS direction, we take cars coming from south to north into account. The superscript "+" means the car is in the intersection, while "↑" means the car is in the next section. For example, if we take the traffic light of EW direction, V_T^+ is the speed of vehicles in the intersection in EW direction, and $V_O^↑$ is the speed of vehicles running out of the intersection from south to north. Empty(O_In) means that there is no coming car in the opposite direction. O_light stands for the traffic in the other direction. In addition, the unit of speed is kilometers per hour (km/h). And we set V_{max} as 40 km/h and V_{min} as 20 km/h.

b) *Algorithm Safety and Liveness*: For an algorithm, safety and liveness are two important properties to be considered. In this system, these two properties are defined as follow:

- **Safety** Safety means the situation where both traffic lights are green simultaneously is not allowed. Because this situation may cause traffic chaos and accidents.
- **Liveness** The algorithm allows the appearance that traffic lights in both directions are red. This happens when it is too crowded to let more car running in. Liveness guarantees once this situation occurs, traffic light of one direction can eventually go back to green.

To ensure safety and liveness, the algorithm has made some description. For example, if both lights are red, and next section of one direction is not blocked, then reduce the red time of that direction. Besides, the traffic light needs to obey the following rules.

- **Rule 1** If a red light should be switched, but the other light is green and is not going to change, then the red

Algorithm 1 Traffic Light Control Algorithm

Input:

Speeds of different direction and location

Output:

Light time

```
1: if the light is currently green then
2:   if ( $V_T^+ > V_{max}$  &&  $V_T^+ > V_{max}$  && ( $V_O^+ < V_{min}$ ) then
3:     check whether the remaining time is less than or
       equal to 10 seconds, if so, add 10 seconds to green
       time
4:   end if
5:   if ( $V_T^+ > V_{min}$  &&  $V_T^+ > V_{min}$  && !Empty(T_In) && Empty(O_In)) then
6:     check whether the remaining time is less than or
       equal to 10 seconds, if so, add 10 seconds to green
       time
7:   end if
8:   if  $V_T^+ < V_{min}$  &&  $V_O^+ > V_{min}$  then
9:     check whether the remaining time is greater than or
       equal to 10 seconds, and whether the green light has
       been maintained for at last 30 seconds, if so, turn
       green light to red
10:  end if
11:  if  $V_T^+ < V_{min}$  &&  $V_O^+ > V_{min}$  then
12:    check whether the remaining time is greater than or
       equal to 10 seconds, and whether the green light has
       been maintained for at last 30 seconds, if so, turn
       green light to red
13:  end if
14: end if
15: if the light is currently red then
16:   if ( $V_T^+ < V_{min}$ ) && ( $V_O^+ > V_{max}$  &&  $V_O^+ > V_{max}$ ) then
17:     check whether the remaining time is less than or
       equal to 10 seconds, if so, add 10 seconds to red
       time
18:   end if
19:   if Empty(T_In) && ( $V_O^+ > V_{min}$  &&  $V_O^+ > V_{min}$ ) && !Empty(O_In) then
20:     check whether the remaining time is less than or
       equal to 10 seconds, if so, add 10 seconds to red
       time
21:   end if
22:   if  $V_T^+ > V_{min}$  && O_light==RED then
23:     check whether the remaining time is greater than or
       equal to 10 seconds, and whether the red light has
       been maintained for at last 30 seconds, if so, reduce
       ret time by 10 seconds
24:   end if
25: end if
```

light has to wait until the green light should be switched too. Then two lights can be switched at the same time.

- **Rule 2** If both traffic lights are red, and happen to be switched at the same moment, then switch the light which has been kept red for longer time.

c) *Algorithm Simulation:* To verify whether the algorithm is correct, we simulate the algorithm with Simulink/MATLAB. We construct a simulation model shown in Fig. 3. This model describes how the algorithm works. The system has three states, R_G, G_R and R_R. The anterior letter represents the traffic light of EW direction, while the posterior one is of NS direction. The initial state is R_G. At every second, the function *random()* generates four random numbers between 0-70 for V_{c_EW} (stands for V_{EW}^+), V_{n_EW} (stands for V_{EW}^+), V_{c_NS} (stands for V_{NS}^+), V_{n_NS} (stands for V_{NS}^+), and two random integers with value 1 or value 0 for Empty_EW_in (stands for Empty(EW_In)) and Empty_NS_in (stands for Empty(NS_In)). If Empty_EW_in is assigned with value 1, then there is no coming car in EW direction, and value 0 means there are some cars coming in. The model receives these numbers as input, and then sets interval time. If the traffic light in EW direction is green, the interval time EW_int takes the return value of function *EW_green_p()*. Function *EW_green_p()* is defined in Function.1. The definition of other functions, namely *EW_red_p()*, *NS_green_p()* and *NS_red_p()*, are similar to that. So we omit those definitions. The variable EW_clock and NS_clock represent how long has the light been kept. If clock time equals to interval time, the traffic lights make the corresponding transitions between states.

Function 1: EW_green_p()

```
1 function interval=EW_green_p()
2 if ( $V_{c\_EW} > MAX$  &&  $V_{n\_EW} > MAX$  && NS_light==RED &&
    $V_{n\_NS} < MIN$ )
3   if (EW_int <= EW_clock+10)
4     interval = EW_int+10;
5   else
6     interval = EW_int;
7   end
8 elseif ( $V_{c\_EW} > MIN$  &&  $V_{n\_EW} > MIN$  && Empty_EW_in==0 &&
   (NS_light==RED) && Empty_NS_in==1)
9   if (EW_int <= EW_clock+10)
10    interval = EW_int+10;
11  else
12    interval = EW_int;
13  end
14 elseif ( $V_{c\_EW} < MIN$  &&  $V_{n\_NS} > MIN$ )
15   if (EW_int >= EW_clock+10 && EW_clock > 30)
16     interval = EW_clock;
17   else
18     interval = EW_int;
19   end
20 elseif ( $V_{n\_EW} < MIN$  &&  $V_{n\_NS} > MIN$ )
21   if (EW_int >= EW_clock+10 && EW_clock > 30)
22     interval = EW_clock;
23   else
24     interval = EW_int;
25   end
26 else
27   interval = EW_int;
28 end
```

We generate the speed of cars randomly for one hour, and then see how the traffic light works. The random speed as input is shown in Fig. 4. From up to down, the sub-graphs represent how V_{EW}^+ , V_{EW}^+ , V_{NS}^+ , V_{NS}^+ , Empty(EW_In) and Empty(NS_In) change during this time. The result of the

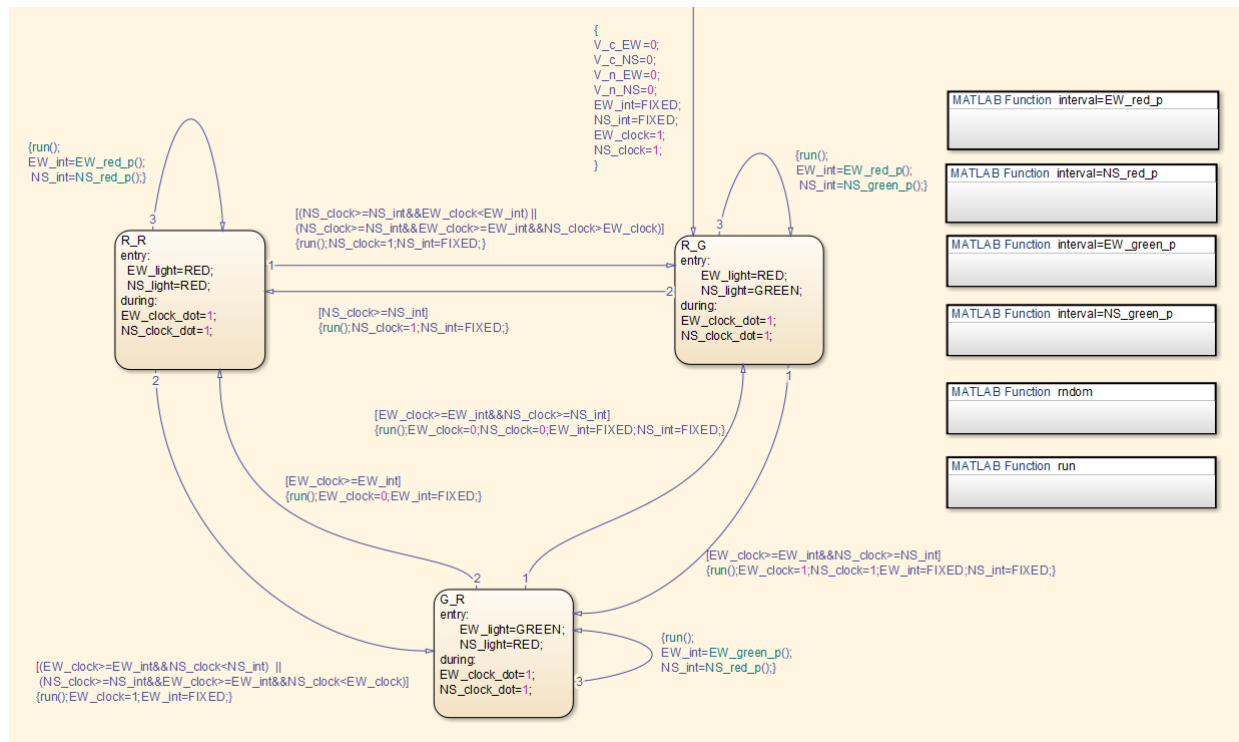


Fig. 3: Simulation Model

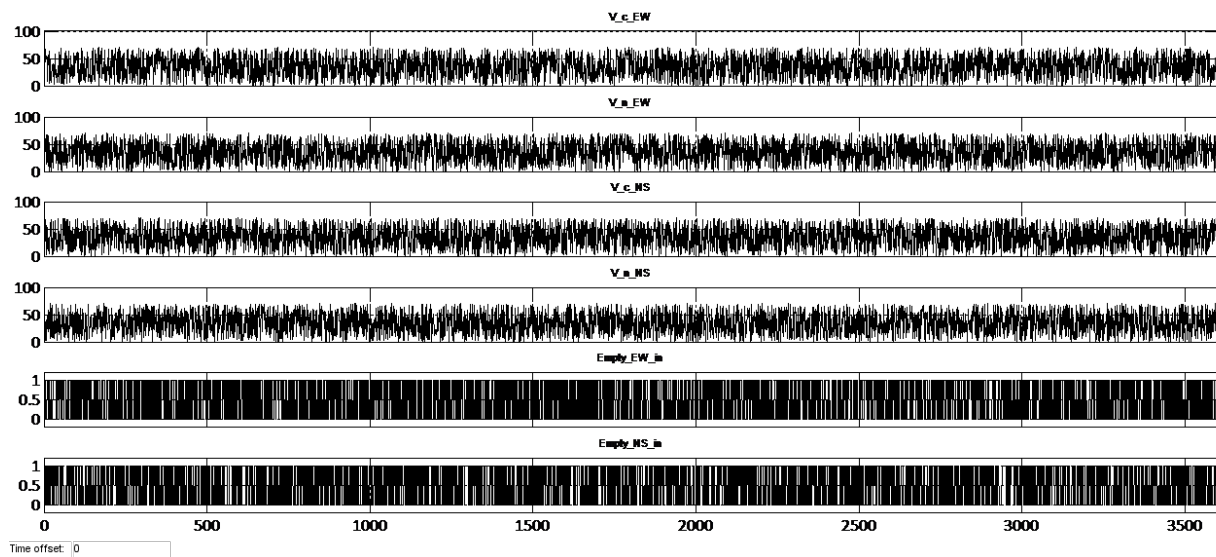


Fig. 4: Random Car Speed

simulation is presented in Fig. 5. In this figure, the red solid line represents the traffic light of EW direction, while the black dotted line stands for the light of NS direction. At every point in time, the light takes one of two values, value 1 or value 0. Value 1 means the light is green, and 0 stands for red.

To make it clear, we pick three regions as three cases among 3600 seconds.

- **Case 1** First, we find at what time the traffic lights is switched for the first time. Fig. 6 shows that, at 32nd second, the traffic light in NS direction becomes red, and the other light is switched at 41st second. At 31nd second, $V_{NS}^\uparrow$ is 13.2269km/h, which means it is blocked. While $V_{EW}^\uparrow$ is 52.1285km/h and is expedite. It meets the condition to be switched from green to red, so the light is switched. For the light in EW direction, after the NS traffic light becomes red, at 32nd second, $V_{EW}^\uparrow$ is 54.6159km/h, is greater than V_{min} , so the red time is reduced by 10 seconds, at 33rd second, $V_{EW}^\uparrow$ is 31.2749km/h, the red time is cut again to 40 seconds, so the light goes to green at 41st second.

Note the period from 32nd second to 41st second, during this time, both lights are red. The system reaches this R_R state because the block of NS direction. Then the red time of unblocked direction is shortened and the system transfer to G_R state.

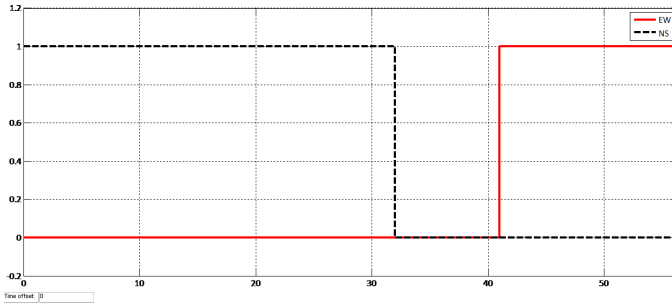


Fig. 6: Case 1: The first time that the traffic lights convert

- **Case 2** This case is where the time of light is lengthened. As Fig. 7 shows, the traffic light in EW direction keeps red for 70 seconds (from 2975rd second to 3045rd second). After analyze the speed, we find that the light time changes 7 times. At 3007th, 3009th, and 3035nd second, the time is shortened, and at 3023th, 3031th, 3034st, and 3038th second, the time is added by 10 seconds. As a result, the final time of this period is 70 seconds.
- **Case 3** From 1740th second to 1801 second, the traffic light of NS direction kept red for 60 seconds. This is the normal situation because the light time is not extended or shortened. At 1801st second, two lights convert at the same time. That is the system transfers from state G_R to state R_G. This case is shown in Fig. 8.

The simulation indicates that this system can control the traffic light dynamically according to the algorithm. Also,

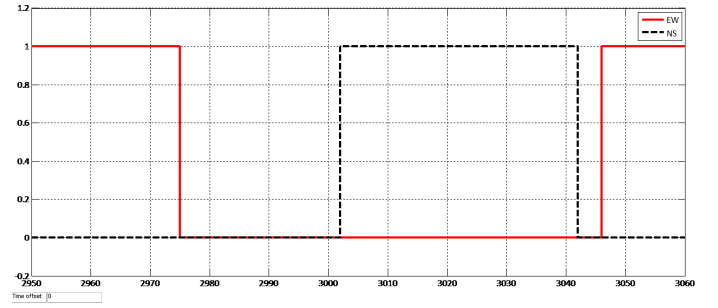


Fig. 7: Case 2: The extended red time of EW direction

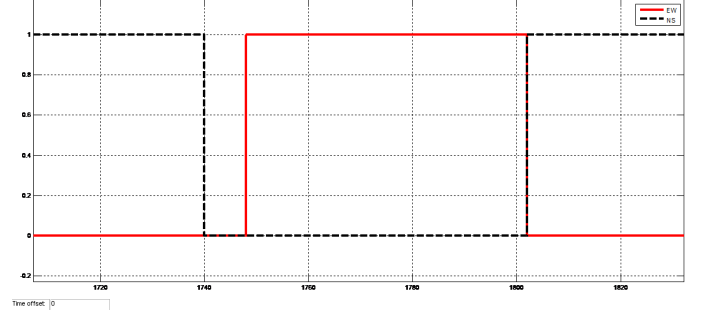


Fig. 8: Case 3: The normal situation

in this simulation, no situation that violates the safety and liveness properties.

IV. RELATED WORK

A. SCOOT

The Split Cycle Offset Optimisation Technique (SCOOT) was developed by the Transport Research Laboratory (TRL) in collaboration with the UK traffic systems industry. It is an adaptive traffic control system which responds automatically to traffic fluctuations. This method of coordination adjusts the signal timings in frequent, small increments to match the latest traffic situation. SCOOT can reduce road delays and has been used in many cities. However, the installation of SCOOT is quite complex and cumbersome.

B. SCATS

The Sydney Coordinated Adaptive Traffic System (SCATS), is an intelligent transportation system developed in Australia in the 1970s. SCATS primarily tries to find the best phasing (i.e. cycle times, phase splits and offsets) for the current traffic situation. This is based on the automatic plan selection from a library in response to the data derived from traffic sensors. The problem of the system is that the plans are fixed, which limits the optimization.

C. Traffic Light Control Systems based on WSN

In this subsection we talk about two systems as representatives of traffic light control systems based on Wireless Sensor Network (WSN).

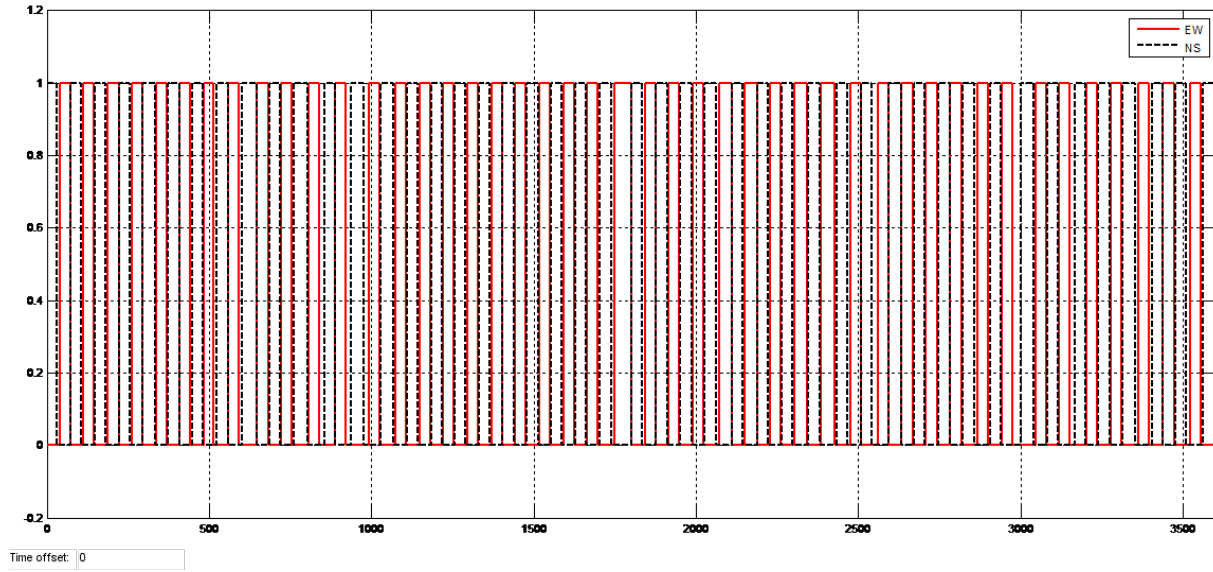


Fig. 5: Result of Simulation.

NOTE: Red solid line: traffic light of EW direction. Black dotted line: traffic light of NS direction.
Value 1: Green. Value 0: Red.

The system proposed in [6] is based on time slices control mechanism. It cuts time into slices, and in each slice it compute the green ratio (the percentage of green time in each time slice). The purpose of the system is to get the maximum average number of passed vehicles in unit time. However, it ignores the average waiting time of vehicles.

[7] introduced an adaptive traffic control system using WSN by building an M/M/1 queue model. This system controls the traffic light based on the optimization of the average waiting time and average queue length. A test bed developed for real measure was presented in this work.

V. CONCLUSION

Traffic congestion is one of the most serious problems now. In order to raise the road capacity and alleviate the traffic congestion, we propose a self-adaptive traffic light control system by using ZigBee, OBD, and other techniques. In this system, when coming into the system range, vehicles will collect and send speed information to the traffic light controller. The controller receives the information and controls the traffic lights according to the speed. A traffic light control algorithm is proposed in this paper. And the simulation verifies the correctness of this algorithm. Compared with the original fixed time traffic light control system, this system can effectively decrease the congestion and raise the passing capacity of the road intersections. If this kind of SATL control systems are constructed in the road then the transportation system are more intelligent and smooth.

The existing shortcomings at present is that the turning of cars are not taken into consideration. So, further refinement of the system is needed.

ACKNOWLEDGMENT

This work is supported by National Natural Science Foundation of China(No.61370100), Shanghai Knowledge Service Platform Project(No.ZF1213), and Project supported by the Shanghai Committee of Science and Technology(No.14511100400).

REFERENCES

- [1] Masaki I, Demir R, Crawley E F. System design for intelligent transportation systems[C] *Intelligent Vehicles Symposium, 1996. Proceedings of the 1996 IEEE*. IEEE, 323-326, 1996.
- [2] Na Zhao, Jiabin Yuan, Han Xu. Survey on Intelligent Transportation System[J]. *Computer Science*, 41(11), 7-11, 2014.
- [3] Jinliang Zhao. The Design and Realization of Self-Adaptive Crossing Traffic Control System[J]. *Journal of Taiyuan University of Technology*, 44(4), 35-355, 2013.
- [4] Hunt P. B., Robertson D. I., Bretherton R. D., et al. SCOOT - A Traffic Responsive Method of Coordinating Signals[J]. *Publication of Transport & Road Research Laboratory*, 1981.
- [5] Long Chen, Sendi Peng, Ming Wei. Study on optimal timing scheme of SCATS[J]. *Silicon Valley*, (12), 186, 2010.
- [6] Qingquan Li, Binluo Liu. Research on traffic light control system based on Wireless Sensor Network[J]. *Science & Technology Association Forum*, (6), 71-73, 2010.
- [7] Khalil M. Yousef, Jamal N. Al-karaki, Ali M. Shatnawi. Intelligent Traffic Light Flow Control System Using Wireless Sensors Networks[J]. *Journal of information science and engineering*, 26(3), 753-768, 2010.
- [8] Wenbin Ye. Design and simulation for real-time traffic flow control model research based on traffic light[D]. *East China Normal University, Shanghai*, 2015.
- [9] Yiqing Song, Run Zhang. Design and Realization of Self-Adaptive Traffic Light Control System[J]. *Computer Measurement & Control*, 16(4), 497-499, 2008.
- [10] Milanes V, Villagra J, Godoy J, et al. An Intelligent V2I-Based Traffic Management System[J]. *IEEE Transactions on Intelligent Transportation Systems*, 13(1), 49-58., 2012.
- [11] Yun Wei, Huaiwei Lu, Zhaohui He. Research on V2I communication performance of intelligent traffic system based on OPNET[J]. *Automation & Instrumentation*, (2), 8-10, 2015.

- [12] Jianqiang Wang, Chenwen Wu, Xiaojun Li. Research on Architecture and Key Technologies of Internet of Vehicles[J]. *Control & Automation*, 27(4), 156-158, 2011.
- [13] Long SunMengliang LiDa Xu. The Application and Development of OBD Technology[J]. *Auto Engineer*, (10), 54-58, 2011.
- [14] Hongquan Pu, Junying Jia, Xiaojiao Zhang, Jianwei Sun. A Survey on ZigBee Network Technology Reaearch[J]. *Computer Systems & Applications*, 22(9), 6-11, 2013.